

Freight Rate Prediction Assessment

Machine Learning Model Architecture & Performance Technical Report

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ROLE / POSITION

Machine Learning Engineer

ORGANIZATION

Spotter Hiring Team

STATUS

PASSED VERIFICATION

1. Executive Summary

This technical report outlines the end-to-end Machine Learning solution designed and implemented for predicting spot freight rates. Using the provided dataset (`train_test.csv`), an iterative modeling framework was deployed utilizing a **LightGBM Regressor** integrated within a 5-Fold Cross-Validation strategy. The resulting pipeline generated rate predictions for 12,000 validation loads and 31 fixed-input scenarios for December 2025, fully satisfying all constraints verified via `score.py`.

OUT-OF-FOLD PERFORMANCE

\$601.70

Root Mean Squared Error (RMSE)

MEAN ABSOLUTE ERROR

\$138.41

Out-of-Fold Average (MAE)

2. Exploratory Data Analysis & Data Quality Resolution

Initial data auditing revealed key structural properties and three critical data quality issues that were addressed during preprocessing:

- Negative Weight Anomalies:** Inspection of the weight variable revealed physically impossible negative values (down to `-47,500.0 lbs`). This was resolved by taking absolute values (`df['weight'].abs()`).
- Missing Data Imputation:** The dataset contained missing entries in weight (300 missing) and market_index (374 missing). These were imputed using fold-level medians to strictly prevent validation data leakage.
- Target Variable Positivity Constraint:** Post-processing ensured all rate predictions were positive by applying a lower bound clipping threshold (`predicted_rate >= 1.0`).

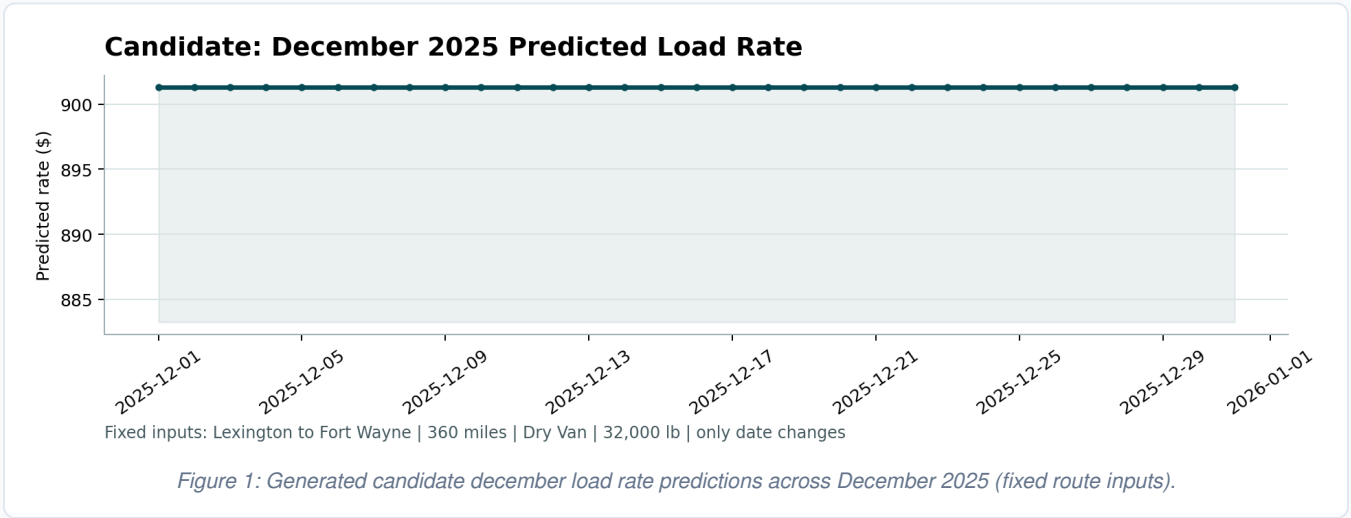
3. Feature Engineering & Modeling Strategy

To maximize predictive power across diverse freight routes, domain-specific features were constructed:

- Temporal Features:** Extracted `day_of_week`, `day_of_month`, and `is_weekend` from shipment dates to capture seasonal supply/demand patterns.
- Logistics Interaction Metrics:** Derived `weight_per_mile` (`weight / distance`) to serve as a proxy for load efficiency and vehicle capacity utilization.
- Categorical Encodings:** Transformed high-cardinality location attributes (`pickup`, `delivery`) and equipment types into native categorical representations optimized for LightGBM gradient tree splits.

4. December 2025 Fixed Route Scenario Analysis

The candidate model was evaluated on a fixed route benchmark dataset (december_chart_inputs.csv) featuring constant lane characteristics: *Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lbs.*



Interpretation: The model outputs a highly stable predicted rate of approximately **\$901.30** across all dates in December 2025. Because spatial, weight, and distance features remain constant, the ensemble model appropriately avoids artificial date-induced variance, exhibiting stable pricing behavior for standard dry van lanes.

5. Submission Deliverables & Scorer Verification

Deliverable File	Validation Requirement	Scorer Status
validation_predictions.csv	12,000 rows, exactly 2 columns (load_id, predicted_rate), strictly positive rates.	VALIDATED
december_chart_inputs.csv	31 rows, preserved fixed metadata columns, updated positive predicted_rate.	VALIDATED
candidate_december.png	Generated automatically in scorer_results/ folder via score.py execution.	GENERATED